$$\begin{array}{r|cc|}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#2}_{0^+} \\ \hline
\mathcal{K}^{\#1}_{0^+}{}^{\dagger} & \Upsilon_1 k^2 + \overset{(2)}{\kappa}_2 & \frac{1}{2}(\Upsilon_2 k^2 + 2\,\overset{(2)}{\kappa}_2) \\
\mathcal{K}^{\#2}_{0^+}{}^{\dagger} & \frac{1}{2}(\Upsilon_2 k^2 + 2\,\overset{(2)}{\kappa}_2) & \Upsilon_3 k^2 + \overset{(2)}{\kappa}_2 \\ \hline
\end{array}$$

$$\begin{array}{r|cc|}
 & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\ \hline
\mathcal{K}^{\#1}_{1^-}{}^{\dagger\alpha} & \frac{1}{3}(\Upsilon_4 k^2 + \overset{(2)}{\kappa}_2) & \frac{1}{6}(\sqrt{5}\,\Upsilon_5 k^2 + 2\sqrt{5}\,\overset{(2)}{\kappa}_2) \\
\mathcal{K}^{\#2}_{1^-}{}^{\dagger\alpha} & \frac{1}{6}(\sqrt{5}\,\Upsilon_5 k^2 + 2\sqrt{5}\,\overset{(2)}{\kappa}_2) & \frac{1}{3}(5 k^2\,\overset{(4)}{\kappa}_2 + 5\,\overset{(2)}{\kappa}_2) \\ \hline
\end{array}$$

$$\begin{array}{r|c|}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} \\ \hline
\mathcal{K}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & 0 \\ \hline
\end{array}
\qquad
\begin{array}{r|c|}
 & \mathcal{K}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\ \hline
\mathcal{K}^{\#1}_{3^-}{}^{\dagger\alpha\beta\chi} & 0 \\ \hline
\end{array}$$

$$\begin{array}{r|cc|}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#2}_{0^+} \\ \hline
\mathcal{J}^{\#1}_{0^+}{}^{\dagger} & \frac{\Upsilon_3 k^2 + \overset{(2)}{\kappa}_2}{\mathrm{Det}(0^+)} & \frac{-\Upsilon_2 k^2 - 2\,\overset{(2)}{\kappa}_2}{2\,\mathrm{Det}(0^+)} \\
\mathcal{J}^{\#2}_{0^+}{}^{\dagger} & \frac{-\Upsilon_2 k^2 - 2\,\overset{(2)}{\kappa}_2}{2\,\mathrm{Det}(0^+)} & \frac{\Upsilon_1 k^2 + \overset{(2)}{\kappa}_2}{\mathrm{Det}(0^+)} \\ \hline
\end{array}$$

$$\begin{array}{r|cc|}
 & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\ \hline
\mathcal{J}^{\#1}_{1^-}{}^{\dagger\alpha} & \frac{5 k^2\,\overset{(4)}{\kappa}_2 + 5\,\overset{(2)}{\kappa}_2}{3\,\mathrm{Det}(1^-)} & \frac{-\sqrt{5}\,\Upsilon_5 k^2 - 2\sqrt{5}\,\overset{(2)}{\kappa}_2}{6\,\mathrm{Det}(1^-)} \\
\mathcal{J}^{\#2}_{1^-}{}^{\dagger\alpha} & \frac{-\sqrt{5}\,\Upsilon_5 k^2 - 2\sqrt{5}\,\overset{(2)}{\kappa}_2}{6\,\mathrm{Det}(1^-)} & \frac{\Upsilon_4 k^2 + \overset{(2)}{\kappa}_2}{3\,\mathrm{Det}(1^-)} \\ \hline
\end{array}$$

$$\begin{array}{r|c|}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} \\ \hline
\mathcal{J}^{\#1}_{2^+}{}^{\dagger\alpha\beta} & 0 \\ \hline
\end{array}
\qquad
\begin{array}{r|c|}
 & \mathcal{J}^{\#1}_{3^-}{}_{\alpha\beta\chi} \\ \hline
\mathcal{J}^{\#1}_{3^-}{}^{\dagger\alpha\beta\chi} & 0 \\ \hline
\end{array}$$

Abbreviations used in matrices

$$\Upsilon_1 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_4 \;\&\&\; \Upsilon_2 == 2\,(\overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2) + \overset{(4)}{\kappa}_4 \;\&\&\; \Upsilon_3 == \overset{(4)}{\kappa}_1 + \overset{(4)}{\kappa}_2 \;\&\&$$

$$\Upsilon_4 == \overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_4 \;\&\&\; \Upsilon_5 == 2\,\overset{(4)}{\kappa}_2 + \overset{(4)}{\kappa}_4 \;\&\&\; \mathrm{Det}(0^+) == -\frac{1}{4} k^4\,\overset{(4)}{\kappa}_4{}^2 \;\&\&\; \mathrm{Det}(1^-) == -\frac{5}{36} k^4\,\overset{(4)}{\kappa}_4{}^2$$

Lagrangian

$$\overset{(2)}{\kappa}_2\,\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\mathcal{K}_{\beta}{}^{\chi}{}_{\chi} + \overset{(4)}{\kappa}_1\,\partial_{\beta}\mathcal{K}_{\chi}{}^{\delta}{}_{\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_2\,\partial_{\chi}\mathcal{K}_{\beta}{}^{\delta}{}_{\delta}\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_4\,\partial^{\chi}\mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\,\partial_{\delta}\mathcal{K}_{\beta\chi}{}^{\delta}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\,\mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{2^+}{}^{\alpha\beta} == 0$	5	$k^{\chi}\,(\partial_{\epsilon}\partial_{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 2\,\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\chi}{}^{\delta\epsilon} - 3\,\partial_{\epsilon}\partial_{\delta}\partial_{\chi}\partial^{\alpha}\mathcal{J}^{\beta\delta\epsilon} - 3\,\partial_{\epsilon}\partial_{\delta}\partial_{\chi}\partial^{\beta}\mathcal{J}^{\alpha\delta\epsilon} + 3\,\partial_{\epsilon}\partial^{\epsilon}\partial_{\delta}\partial_{\chi}\mathcal{J}^{\alpha\beta\delta} + \eta^{\alpha\beta}\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial_{\chi}\mathcal{J}^{\delta\epsilon\phi} - \eta^{\alpha\beta}\,\partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\chi}\mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon}) == 0$
$\mathcal{J}^{\#1}_{3^-}{}^{\alpha\beta\chi} == 0$	7	$k^{\delta}\,k^{\epsilon}\,(3\,\partial_{\epsilon}\partial^{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\delta}{}^{\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\beta\phi}{}_{\phi} - \partial_{\epsilon}\partial_{\delta}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\alpha\phi}{}_{\phi} + 2\,\partial_{\phi}\partial^{\chi}\partial^{\beta}\partial^{\alpha}\mathcal{J}_{\delta\epsilon}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\beta}\partial^{\alpha}\mathcal{J}^{\chi}{}_{\delta}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\partial^{\alpha}\mathcal{J}^{\beta}{}_{\delta}{}^{\phi} - 4\,\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\partial^{\beta}\mathcal{J}^{\alpha}{}_{\delta}{}^{\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\beta\chi\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\alpha\chi\phi} + 5\,\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\alpha\beta\phi} - 5\,\partial_{\phi}\partial^{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\beta\chi} - \eta^{\beta\chi}\,\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\alpha}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} - \eta^{\alpha\chi}\,\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\beta}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} - \eta^{\alpha\beta}\,\partial_{\gamma}\partial_{\epsilon}\partial_{\delta}\partial^{\chi}\mathcal{J}^{\phi}{}_{\phi}{}^{\gamma} + \eta^{\beta\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\alpha}\mathcal{J}_{\delta}{}^{\phi\gamma} + \eta^{\alpha\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\beta}\mathcal{J}_{\delta}{}^{\phi\gamma} + \eta^{\alpha\beta}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial^{\chi}\mathcal{J}_{\delta}{}^{\phi\gamma} - \eta^{\beta\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\phi\gamma} - \eta^{\alpha\chi}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\beta\phi\gamma} - \eta^{\alpha\beta}\,\partial_{\gamma}\partial_{\phi}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\phi\gamma} + \eta^{\beta\chi}\,\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\alpha\phi}{}_{\phi} + \eta^{\alpha\chi}\,\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\beta\phi}{}_{\phi} + \eta^{\alpha\beta}\,\partial_{\gamma}\partial^{\gamma}\partial_{\epsilon}\partial_{\delta}\mathcal{J}^{\chi\phi}{}_{\phi}) == 0$
Total # constraint(s):	12	
Resolved unitarity condition(s):		True